A System-Driven Taxonomy of Foundation Models for Time Series Classification

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Abstract—Foundation models (FMs) have arrived in the time series domain with remarkable speed—but overwhelmingly in pursuit of forecasting, and the surveys that map this literature treat classification as one downstream task among many. We argue that time series classification (TSC) deserves analysis on its own terms. Where forecasting is generative, classification is discriminative, and its binding constraints are different: a pre-trained representation must be reduced to a label under severe label scarcity, and the model must run wherever the decision is made—from cloud services to embedded sensors and wearables. These are questions of *adaptation* and *deployment*, not merely of how a model was pre-trained. We present a *system-driven* account of foundation models for TSC, organized around the four groups that specify any such model—*design* (what it learns from and how it is built), *scale*, *interface* (how data enter the model and how a decision is produced), and *deployment*; read as a taxonomy, these groups surface the choices that decide whether a model can serve as a classifier at all. Within the design group we isolate a well-defined, three-level *spine*—source modality → pre-training objective → architecture—and use it to place more than forty representative models on a single map. The shape of the field then becomes a finding in itself: pre-training is heavily concentrated in time-series-native models; Transformers dominate the architecture space, with the few convolutional, recurrent, and MLP-mixer alternatives clustered at the compact end; adaptation for classification lags well behind forecasting; and, most tellingly, compact models such as TSPulse now match or surpass far larger ones on classification—evidence that the field's reflexive emphasis on scale is partly misplaced for TSC. We close with a system-driven decision framework that maps a problem's constraints—input dimensionality, label budget, and hardware target—directly onto viable regions of the taxonomy. This is a position-and-survey paper: its contribution is the organizing framework, not new empirical results.

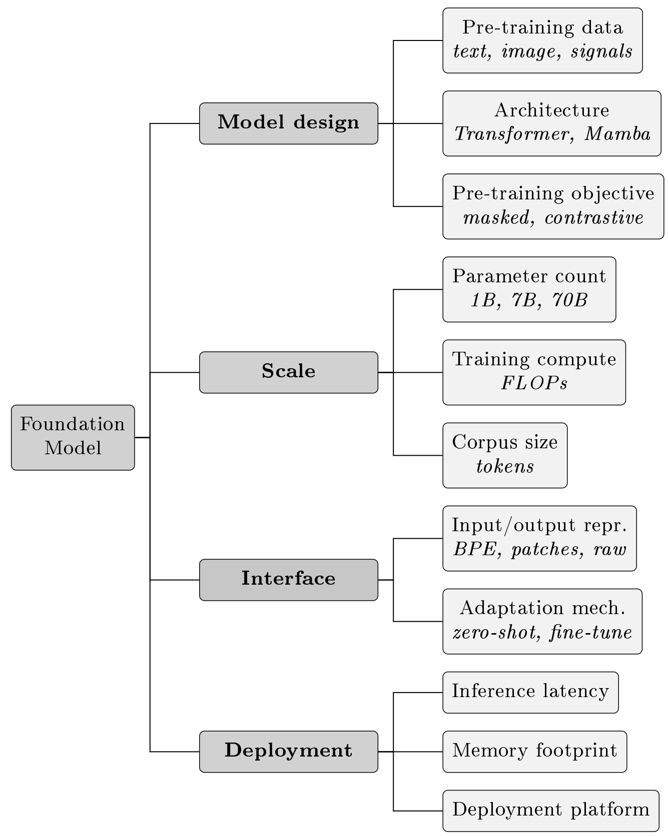

Figure 1: The organizing framework of this paper. A foundation model is specified by four groups—model design, scale, interface, and deployment—which we use as the structure of the paper. The italic line under each leaf is an illustrative value, not an exhaustive list; Sections 3–6 take each group in turn, read as a taxonomy dimension for TSC.

1 Introduction

Time series classification (TSC) is a foundational problem across healthcare (ECG and EEG diagnosis), human activity recognition, audio and speech, and industrial condition monitoring. It is one of several time-series tasks—alongside forecasting, regression, anomaly detection, and imputation—but recent foundation-model work has concentrated on forecasting. Classification differs from forecasting in ways that bear directly on model design: forecasting is *generative*, predicting future values from past ones, whereas classification is *discriminative*, mapping an entire series to a categorical label. The two also differ in their data economics—labels in TSC are scarce and expensive, often requiring expert annotation—and in their canonical benchmarks, the UCR univariate archive [4] and the UEA multivariate archive [5].

Foundation models transformed natural language processing and computer vision by decoupling a large, self-supervised pre-training phase from inexpensive downstream adaptation [1]. This paradigm has now reached time series, but predominantly through forecasting.

A substantial body of recent models—TimeGPT [14], TimesFM [8], Chronos [7], Moirai [9], Lag-Llama [10], Timer [11], and Time-MoE [12]—targets zero-shot or few-shot forecasting. Classification has remained a secondary concern: it appears as one task among many in general-purpose models such as GPT4TS [15] and UniTS [13], but rarely as the central design objective.

The surveys that organize this space reflect the same emphasis. Liang et al. [2] provide a broad tutorial-and-survey of foundation models for time series *analysis*, organized largely by architecture, pre-training, and modality. Ye et al. [3] center the large-language-model route to time series representation. Both are valuable and broad, but both are organized around how models are *built* and treat classification as one of several interchangeable downstream tasks. Neither is written from the standpoint of a practitioner who has a specific classification problem, a specific data shape, and a specific hardware budget.

We adopt this practitioner standpoint, which we term *system-driven*. Our central claim is that for TSC specifically, the questions of primary importance extend beyond how a model was pre-trained to whether it can ingest a multivariate, variable-length signal, whether it can be adapted from the few labels typically available, and whether it can run on the device where classification must occur. These are capability and deployment questions, and a taxonomy for TSC should treat them as primary concerns.

Contributions.

- We organize the specification of a foundation model into four groups—design, scale, interface, and deployment (Figure 1)—and use these groups as the structure of the paper.
- We read each group as a taxonomy for TSC (Table 1), detailing the dimensions in each that decide whether a model is usable as a classifier.
- Within the design group we isolate a well-defined, three-level *spine*—source modality → pre-training objective → architecture—and use it to place more than forty models (Figure 2, Table 2).
- We give a gap analysis organized by the four groups and a deployment-oriented decision view (Section 7).

2 Background

2.1 Time series classification

A univariate time series is an ordered sequence $x_{1:T}$; a multivariate series stacks C channels, $x_{1:T}^{(1:C)}$. TSC assigns a series to one of K classes. Three properties make the problem distinctive for foundation models. First, *label scarcity*: many UCR/UEA datasets contain only tens to hundreds of labeled examples; few-shot and zero-shot regimes are therefore the norm rather than the exception. Second, *heterogeneous shape*: series lengths vary by orders of magnitude across datasets, channel counts differ, and sampling can be irregular—so whether a model natively handles multivariate or variable-length input is decisive. Third, the *output is a label, not a horizon*, which means the model's representation must be reduced to a class decision, foregrounding the adaptation mechanism.

2.2 Foundation models

Following Bommasani et al. [1], a foundation model is trained on broad data, typically self-supervised, at scale, and adapted to many downstream tasks. The defining characteristic is the separation of an expensive, general pre-training phase from inexpensive task adaptation. The specification of any such model can be summarized by four groups of properties, shown in Figure 1: its *design* (the data it learns from, its architecture, and its pre-training objective), its *scale* (parameters, compute, corpus size), its *interface* (how inputs and outputs are represented, and how it is adapted), and its *deployment* profile (latency, memory, target platform). We treat provenance and licensing as metadata rather than specification, since they describe a model without predicting its behavior. These four groups—design, scale, interface, and deployment—organize the rest of this paper (Sections 3–6), and Table 1 reads each group as a taxonomy dimension for TSC.

2.3 Classification and generative pre-training

The pre-training objectives that dominate time-series FMs—masked reconstruction and autoregressive generation—are optimized for predicting values rather than separating classes. A model may be an effective forecaster yet a poor classifier if its representation entangles the factors on which a label depends. Consequently, for TSC the *adaptation* step (the mechanism by which a frozen or fine-tuned representation is converted into a decision) and the *label budget* it requires are not incidental; they are decisive for classification quality. This also explains why contrastive and self-distillation objectives, which explicitly shape a discriminative embedding space, remain competitive for TSC even against substantially larger generative models.

3 Model Design

The first group of Figure 1 concerns what a model learns from, how it is constructed, and how it is trained. For time series classification these three design choices—source modality and pre-training data, pre-training objective, and architecture—most strongly determine the representation a model offers a downstream classifier. They are also where research is most active; we therefore

Group	Dimension	Values / range
Model design	Source modality & data	time-series-native, language-transferred, vision-transferred, multimodal; general / domain-specific / synthetic
	Pre-training objective	masked, autoregressive, reconstruction (seq2seq), contrastive, transformation prediction, self-distillation, supervised multi-task
	Architecture	Transformer (enc / dec / enc-dec), recurrent (RNN / LSTM), state-space (Mamba), conv / MLP-mixer, mixture-of-experts
Scale	Parameters	S (<10M), M (10–100M), L (>100M); for MoE, total vs. active
	Compute & corpus	pre-training FLOPs; corpus size in series / tokens
Interface	Input representation	raw / point-wise, patch, quantized tokens, spectral, image-rendered
	Channel & length	univariate, channel-independent, channel-mixing, variable-length / irregular
	Adaptation for TSC	zero-shot, linear probe, full fine-tune, prompt / in-context
Deployment	Latency & memory	per-series latency, peak memory, quantization (GPU vs. GPU-free)
	Target platform	cloud GPU, edge / embedded, wearable / microcontroller

Table 1: The foundation-model specification of Figure 1, read as a taxonomy for TSC. The four groups—design, scale, interface, deployment—organize Sections 3–6. Source modality and pre-training objective are kept as separate design dimensions because they are orthogonal, which allows boundary cases such as Chronos to be placed unambiguously.

treat this group at greatest length and use it to survey the models (Table 2).

3.1 Source modality and pre-training data

Source modality concerns the origin of a model's prior knowledge. Time-series-native models pre-train on time series directly. Language-transferred models adapt a pre-trained large language model, either reprogramming its input space or freezing its blocks. Vision-transferred models render series as images and reuse a vision backbone. Multimodal models consume the signal jointly with text or other modalities. The data itself ranges from general multi-domain corpora to domain-specific pre-training (ECG, EEG, human activity) and synthetic augmentation; scope matters for TSC because a classifier inherits whatever invariances the pre-training distribution induced. Domain-specific corpora have recently reached very large scale: the wearable foundation models LSM [37] and SensorLM [36] pre-train on tens of millions of hours of smartwatch sensor data.

3.2 Pre-training objective

Following the self-supervised learning literature, we separate generative/reconstructive objectives—masked modeling, autoregressive prediction, and sequence-to-sequence reconstruction—from contrastive objectives, from predictive pretext tasks such as transformation prediction, from self-distillation (non-contrastive teacher–student), and from supervised multi-task training. This split is the one that matters most for classification (Section 2.3): the discriminative objectives—contrastive and self-distillation—shape a label-separating representation directly, whereas the generative ones do so only as a by-product. A recent development enriches masked reconstruction itself: TSPulse [30] augments masked modeling with disentanglement across the time and frequency domains, producing separate fine-grained and semantic embeddings to make zero-shot transfer—including classification—achievable with a compact model.

3.3 Architecture

Transformers dominate, in encoder-only (MOMENT [6], Moirai [9]), decoder-only (TimesFM [8], Timer [11]), and encoder–decoder (Chronos [7]) forms. State-space models [33] and mixture-of-experts (Time-MoE [12]) are emerging at the large end, while convolutional, recurrent, and MLP-mixer backbones persist—and increasingly dominate—at the lightweight end: the contrastive family uses convolutional encoders, the earliest self-supervised wearable models are convolutional (TPN [40]) or recurrent (Activity2Vec [42]), and the compact TSMixer-lineage models TTM [31] and TSPulse [30] are MLP-mixers rather than Transformers. Architecture is loosely correlated with representation (patches with Transformers; raw or spectral inputs with mixers and convolutions), but the two are distinct choices.

3.4 The design spine and a survey of models

The three design dimensions compose into a well-defined hierarchy that we use as the backbone of the survey: source modality → pre-training objective → architecture. These are orthogonal, so a model is a (modality, objective, architecture) triple, and the cross-product is informative precisely because it is not uniformly populated. Separating modality from objective also resolves cases that a flatter taxonomy cannot represent. Chronos [7] is the canonical example: it borrows a language-model *architecture* (T5, encoder–decoder) and a quantized-token *representation*, yet is trained from scratch on *time series* with an autoregressive objective. Fusing "architecture" and "modality" cannot place it; the spine labels it unambiguously as time-series-native × autoregressive × encoder–decoder. Figure 2 depicts the spine with representative models at the leaves, omitting every combination that the literature has not instantiated.

We examine the spine branch by branch; Table 2 places each model on all four specification groups.

Time-series-native: masked. MOMENT [6] is an encoder-only masked model with linear-probing and fine-tuning interfaces that reports classification among its tasks. Moirai [9] is an encoder-only masked forecaster with any-variate support. The transformer framework of Zerveas et al. [32] (TST) established masked pre-training for multivariate representation and remains a classification baseline. Departing from Transformers, TSPulse [30] is a 1M-parameter MLP-mixer whose dual-space (time and frequency) masked reconstruction reaches state-of-the-art multivariate classification on the UEA archive while running without a GPU, outperforming models 10–100× larger. LSM [37], a Vision Transformer masked autoencoder trained on over 40 million hours of wearable sensor data, applies the same masked-reconstruction principle at scale and supports activity recognition as a downstream task.

Time-series-native: autoregressive. Timer [11] and TimesFM [8] are decoder-only generative models; Lag-Llama [10] adds lag-token conditioning; Time-MoE [12] scales via mixture-of-experts; Chronos [7] reframes forecasting as language modeling over quantized tokens. This family is oriented primarily toward forecasting, and its relevance to TSC is through transferable encoder representations rather than native classification heads. The closely related TTM [31] is a compact mixer pre-trained for forecasting that can also be fine-tuned for classification.

Time-series-native: contrastive. This is the most directly classification-oriented family. TS2Vec [21] learns hierarchical contrastive representations and is evaluated on 125 UCR and 29 UEA datasets with a linear classifier on frozen features. TS-TCC [23] contrasts temporal and contextual views; TF-C [24] enforces time–frequency consistency for transfer; CoST [22] disentangles seasonal and trend components. A recent wearable foundation model, Custom HAR-FM [44] (Qiu et al.), combines masked modeling with a contrastive objective in the wav2vec2 style and is designed to absorb heterogeneous HAR datasets that differ in modality, device, and sampling rate. MANTIS [29] (Feofanov et al.) applies a contrastive (InfoNCE-style) objective to a lightweight Vision-Transformer backbone pre-trained on millions of series, and is built for calibrated classification by non-expert users on the edge tier.

Time-series-native: early self-supervised HAR. Before foundation-model terminology, three self-supervised representation learners established the wearable-sensing lineage that the large models (LSM [37], SensorLM [36]) later scaled, and they occupy the non-Transformer, lightweight end of the spine. TPN [40] (Saeed et al.) trains a temporal convolutional network on a multi-task transformation-prediction pretext—classifying which augmentation was applied to the signal—and transfers the learned features across datasets; it is the seminal transformation-prediction approach for HAR. This objective was later scaled by SSL-Wearables [43] (Yuan et al.), which pre-trains a deep convolutional network on 700,000 person-days of UK Biobank accelerometry and releases transferable models that generalize across eight benchmark datasets, devices, and cohorts. Masked Recon. [41] (Haresamudram et al.) applies BERT-style masked reconstruction to body-worn sensor streams with a 1.5M-parameter Transformer encoder, an early wearable counterpart to TST. Activity2Vec [42] (Ghods and Cook) learns transferable activity embeddings with a sequence-to-sequence LSTM autoencoder, the only recurrent model in the survey. These models pre-train without labels and adapt through a frozen encoder and a small classification head.

Time-series-native: self-distillation and supervised. A small but distinct line applies non-contrastive self-distillation—teacher–student training in which the teacher is an exponential moving average of the student and no negative pairs are drawn. SDRL [55] (Pieper et al.) adapts data2vec to time series with a convolutional encoder that predicts the teacher's latent representation of a masked input, while UTICA [56] (Moakher et al.) brings DINOv2-style multi-crop self-distillation with patch masking to a Transformer backbone, reaching state-of-the-art accuracy on the UCR and UEA classification archives. On the supervised side, UniTS [13] trains across tasks (including classification) with shared weights and prompt-style task tokens, and TimesNet [25], though a supervised backbone rather than a pre-trained FM, is a common classification reference.

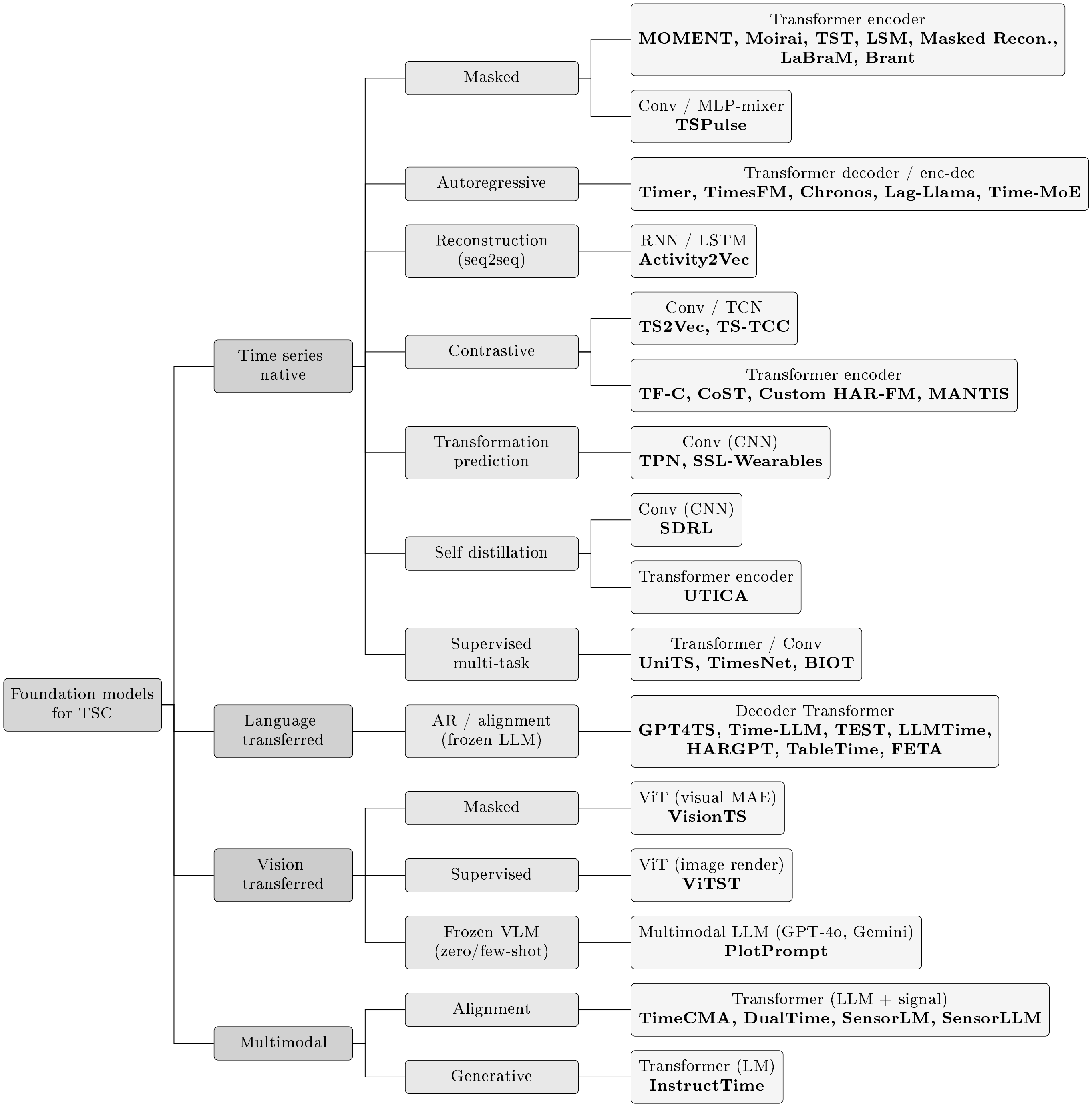

Figure 2: The design spine—source modality → pre-training objective → architecture—with representative models at the leaves. The asymmetry is itself a finding: time-series-native pre-training spans all seven catalogued objectives and is the only branch with non-Transformer architectures—the convolutional contrastive, transformation-prediction, and self-distillation encoders, the recurrent Activity2Vec, and the MLP-mixer TSPulse—whereas the language-, vision-, and multimodal-transferred branches, though increasingly populated, each remain architecturally narrow—uniformly Transformer-based and spanning only one or two objectives.

Time-series-native: EEG and neural-signal models. A distinct domain-specific strand pre-trains on neural recordings for classification under the label scarcity that characterizes clinical EEG. LaBraM [46] (Jiang et al.) trains a vector-quantized neural tokenizer and then a Transformer encoder by predicting masked neural codes over EEG channel patches, fine-tuning for abnormal detection, event-type, and emotion classification across roughly twenty datasets. Brant [48] (Zhang et al.) is a 500M-parameter masked Transformer for intracranial (SEEG) recordings that models temporal and spatial structure jointly and supports seizure detection. BIOT [47] (Yang et al.) instead tokenizes each channel into a "sentence" so that a single model can be pre-trained across datasets with mismatched channels, lengths, and missing values—a direct answer to the channel/length problem—and is fine-tuned for EEG and other biosignal classification. Together these span three orders of magnitude in scale, from the few-million-parameter LaBraM and BIOT to the 500M-parameter Brant, while sharing the cross-dataset, channel-robust design that clinical neural-signal classification requires.

Language-transferred. GPT4TS [15] freezes the self-attention and feed-forward blocks of a language model and adapts only lightweight components, reporting strong classification and few-/zero-shot results. Time-LLM [16] reprograms the LLM input space with text-prototype alignment; TEST [17] aligns time-series embeddings to text prototypes; LLMTime [18] and PromptCast [20] serialize series as text for prompting. HARGPT [19] extends this to classification, prompting a frozen LLM with raw multivariate IMU values to perform zero-shot human activity recognition through chain-of-thought reasoning and no task-specific training. TableTime [51] carries this training-free idea to the multivariate setting, reformulating each series as a table and prompting a frozen LLM to classify it while preserving the channel and temporal structure that flat text serialization discards. FETA [53] pushes the same training-free route further with a multi-agent pipeline that decomposes a multivariate series by channel, retrieves labeled exemplars for each, and has a reasoning LLM compare the query against them before a confidence-weighted vote. Across this branch the architecture is invariably a Transformer inherited from the language backbone, which is why this branch of Figure 2 collapses to a single architecture leaf.

Vision- and multimodal-transferred. VisionTS [27] reuses a visual masked autoencoder as a zero-shot forecaster by rendering series as images; ViTST [28] turns irregularly sampled series into line-plot images and applies a vision Transformer for classification, handling the irregular-sampling case that poses difficulties for many native models. The same image-rendering route extends to general-purpose frozen models: Daswani et al. [39] (PlotPrompt in Table 2) show that rendering a series as a plot and passing it to the vision encoder of an off-the-shelf multimodal model such as GPT-4o or Gemini yields stronger zero- and few-shot classification—demonstrated on inertial fall detection and human activity recognition—than supplying the same series as numerical text, and reduces input-token cost by up to an order of magnitude, without any task-specific training. TimeCMA [34] and DualTime [35] align time series with language, and SensorLM [36] scales this idea to wearable data, pairing roughly 60 million hours of smartwatch signals with text for zero-shot human activity recognition. SensorLLM [38] takes a complementary route to the same modality pairing: rather than pre-training at scale, it aligns a frozen large language model with motion-sensor streams through automatically generated trend descriptions and then attaches a lightweight classifier, matching task-specific baselines on five human-activity-recognition datasets while keeping both the language backbone and the time-series encoder frozen. Departing from the alignment paradigm, InstructTime [52] instead treats classification generatively: it discretizes the series into tokens, performs cross-domain autoregressive pre-training, and fine-tunes a language model to emit the class label as text, so that label semantics and contextual text are modeled jointly rather than collapsed to a one-hot target. This multimodal branch is recent and still sparsely populated, though SensorLM is a large-scale exception; it is where text metadata (e.g. clinical notes alongside an ECG) is most likely to benefit classification.

4 Scale

The second group concerns the size of a model: its parameter count, the compute expended in pre-training, and the size of its corpus. The field spans nearly four orders of magnitude in parameters, from billion-scale forecasters (Time-MoE [12], TimesFM [8]) and frozen large language models (Time-LLM [16], LLMTime [18]) at one end, through the 10–100M-parameter encoders that report classification (MOMENT [6], UniTS [13]), to the 1–10M-parameter models at the other (TS2Vec [21], MANTIS [29], TTM [31], and TSPulse [30]). We report parameters as coarse buckets (Table 2) rather than exact counts, and note that mixture-of-experts models must be read by active rather than total parameters.

For classification specifically, scale and quality are weakly coupled. Scale clearly helps the zero-shot *forecasting* transfer the large native models are built for, but the discriminative signal a classifier requires is shaped more by objective and representation than by parameter count. The clearest evidence is TSPulse [30]: a 1M-parameter model that reports state-of-the-art multivariate classification while outperforming models 10–100× larger on more than seventy datasets. Compute and

Model	Yr	Modality	Objective	Architecture	Repr.	Scale	Multivar.	Adaptation for TSC
TST [32]	’21	TS-native	masked	Transformer enc.	patch/raw	S	yes	probe / fine-tune
TS2Vec [21]	’22	TS-native	contrastive	Conv / TCN	raw	S	yes	linear probe (frozen)
TS-TCC [23]	’21	TS-native	contrastive	Conv + Transformer	raw	S	partial	probe / fine-tune
TF-C [24]	’22	TS-native	contrastive	Transformer enc.	time/freq	S	yes	fine-tune (transfer)
CoST [22]	’22	TS-native	contrastive	Conv / TCN	raw	S	yes	linear probe
TimesNet [25]	’23	TS-native	supervised	Conv (2D-inception)	raw→2D	M	yes	fine-tune
MOMENT [6]	’24	TS-native	masked	Transformer enc.	patch	M	channel-ind.	probe / fine-tune
Moirai [9]	’24	TS-native	masked	Transformer enc.	multi-patch	M	any-variate	zero-shot (fcst.)
LSM [37]	’24	TS-native	masked	ViT (MAE)	patch	L	yes	fine-tune / probe
Timer [11]	’24	TS-native	autoregressive	Transformer dec.	patch	M	univariate	fine-tune
TimesFM [8]	’24	TS-native	autoregressive	Transformer dec.	patch	L	univariate	zero-shot (fcst.)
Chronos [7]	’24	TS-native	autoregressive	T5 enc-dec	quantized	M	univariate	zero-shot (fcst.)
Lag-Llama [10]	’23	TS-native	autoregressive	Transformer dec.	lag tokens	S	univariate	zero-shot / fine-tune
Time-MoE [12]	’24	TS-native	autoregressive	MoE Transformer	patch	L	univariate	zero-shot (fcst.)
MANTIS [29]	’25	TS-native	contrastive	Transformer (light)	patch	S	yes	frozen + head
TTM [31]	’24	TS-native	generative (fcst.)	MLP-mixer	patch	S	yes	fine-tune / probe
TSPulse [30]	’25	TS-native	masked (dual-space)	MLP-mixer	time/freq	S	yes	zero-shot / fine-tune
UniTS [13]	’24	TS-native	supervised	Transformer	patch+prompt	M	yes	prompt / shared head
TPN [40]	’19	TS-native	transf. prediction	Conv (CNN)	raw	S	yes	frozen + head
Masked Recon. [41]	’20	TS-native	masked	Transformer enc.	raw	S	yes	frozen + head
Activity2Vec [42]	’19	TS-native	reconstruction	RNN (LSTM)	raw	S	yes	frozen + head
SSL-Wearables [43]	’24	TS-native	transf. prediction	Conv (ResNet)	raw	S	yes	fine-tune / probe
Custom HAR-FM [44]	’25	TS-native	masked + contrastive	Transformer enc.	raw	S	yes	fine-tune (multi-task)
LaBraM [46]	’24	TS-native	masked	Transformer enc.	patch	S	yes	fine-tune / probe
BIOT [47]	’23	TS-native	supervised (x-data)	Transformer	patch	S	yes	fine-tune / probe
Brant [48]	’23	TS-native	masked	Transformer	patch	L	yes	fine-tune / probe
SDRL [55]	’23	TS-native	self-distillation	Conv (CNN)	raw	S	yes	frozen + head
UTICA [56]	’26	TS-native	self-distillation	Transformer enc.	patch	S	yes	frozen + head
GPT4TS [15]	’23	language	AR (frozen LM)	Transformer (GPT-2)	patch	M	channel-ind.	partial fine-tune
Time-LLM [16]	’24	language	AR (frozen LM)	Transformer (LLM)	reprogram	L	yes	reprogramming
TEST [17]	’24	language	contrastive align	Transformer (LLM)	embeddings	L	yes	prompt / embed
LLMTime [18]	’23	language	AR (frozen LM)	Transformer (LLM)	digit tokens	L	univariate	zero-shot prompt
HARGPT [19]	’24	language	AR (frozen LM)	Transformer (LLM)	raw (text)	L	yes	zero-shot prompt
TableTime [51]	’25	language	AR (frozen LM)	Transformer (LLM)	table / text	L	yes	zero-shot prompt
FETA [53]	’25	language	AR (frozen LM)	LLM (multi-agent)	raw (text)	L	yes	in-context (few-shot)
VisionTS [27]	’24	vision	masked (MAE)	ViT	image	M	univariate	zero-shot (fcst.)
ViTST [28]	’23	vision	supervised	ViT / Swin	image	M	yes (irreg.)	fine-tune
PlotPrompt [39]	’24	vision	frozen VLM	Multimodal LLM	image (plot)	L	yes	zero-/few-shot prompt
TimeCMA [34]	’25	multimodal	alignment	Transformer + LLM	cross-modal	L	yes	alignment / fine-tune
DualTime [35]	’24	multimodal	alignment	LM + adapters	TS / text	L	yes	alignment / fine-tune
SensorLM [36]	’25	multimodal	alignment (contr.+gen.)	Transformer	sensor / text	L	yes	zero-shot / few-shot
SensorLLM [38]	’24	multimodal	alignment (gen.)	Transformer + LLM	sensor / text	L	yes	alignment + head
InstructTime [52]	’25	multimodal	AR (generative)	Transformer (LM)	discrete tokens	M	yes	fine-tune (gen.)

Table 2: Representative foundation models mapped onto the specification of Figure 1. The columns correspond to the four groups: modality, objective, and architecture are *design*; the parameter bucket is *scale*; representation, multivariate support, and adaptation are *interface*; and *deployment* tier follows from scale (S-bucket models are edge-deployable, L-bucket models are cloud-tier). Scale is bucketed—S (<10M), M (10–100M), L (>100M) or a frozen large LM—to avoid asserting exact parameter counts. “fcst.” marks models whose released interface targets forecasting, so their use for classification proceeds via transferred representations. Model attributes should be verified against the cited primary sources.

corpus size are correspondingly hard to compare across models—pre-training mixtures differ in domain coverage and are reported inconsistently—so we treat them qualitatively and caution that a “larger corpus” does not reliably predict better classification transfer.

5 Interface

The third group is, for classification, the most consequential, governing how data enter the model and how a decision is produced. It has three dimensions—input representation, channel and length handling, and the adaptation mechanism.

Input representation. Patching [26] treats sub-windows as tokens; value quantization [7] maps real values to a fixed vocabulary; raw/point-wise inputs, spectral (time–frequency) representations, and image rendering complete the set. The representation choice interacts with classification: TSPulse [30] reads each series in both the time and frequency domains, and the vision route (ViTST [28], VisionTS [27]) rendering series as images is attractive exactly when the raw shape is awkward for sequence models.

Channel and length handling. This is the dimension that general surveys most underweight, and the one that frequently determines usability. The UEA benchmark is multivariate and unequal-length, and real signals are frequently irregularly sampled, yet several prominent models are univariate-only. Whether a model is univariate, channel-independent, channel-mixing, or supports irregular sampling (which ViTST [28] targets directly) can determine whether it applies at all. The classification-strong compact models (MANTIS [29], TSPulse [30]) and the any-variate masked models (Moirai [9]) are multivariate by design.

Adaptation for TSC. This is the most consequential dimension. Options range from zero-shot use of a frozen encoder with a nearest-centroid or linear decision, through linear probing, to full fine-tuning, to prompt and in-context adaptation. Because TSC labels are scarce, the low-cost adaptation regimes are most important: TS2Vec [21] classifies from frozen features, TSPulse [30] offers both zero-shot use and a lightweight fine-tuning module for classification, and GPT4TS [15] adapts only a small fraction of its parameters; in the same vein, Xiong et al. [49] attach lightweight time-series adapters to frozen forecasting foundation models such as Chronos to obtain competitive HAR classification. Many large native models, by contrast, expose only a forecasting interface, so using them as classifiers means probing or fine-tuning transferred representations.

6 Deployment

The fourth group concerns where the model runs: its inference latency, memory footprint, and target platform. This group is routinely neglected in model-centric surveys, yet it is decisive for much real-world TSC, which is performed on wearables, embedded sensors, and other resource-constrained devices rather than in the cloud. A billion-parameter forecaster and a 1M-parameter mixer occupy opposite ends of a deployment spectrum—cloud GPU, edge/embedded, and wearable/microcontroller—and quantization and GPU-free execution move a model along it.

The large native and LLM-based models are effectively cloud-tier: their latency and memory put them out of reach of on-device inference. The same holds for foundation models trained on wearable data: LSM [37] (a 110M-parameter Vision Transformer) and SensorLM [36] are cloud-scale despite originating on the watch, and are distinct from the on-device models below. The edge tier, though still smaller, is now substantial and expanding: MANTIS [29] targets classification for non-expert users, and the TSMixer-lineage TTM [31] and TSPulse [30] run with GPU-free inference at roughly 1M parameters. TSPulse, in particular, demonstrates that this tier need not sacrifice accuracy—it matches or exceeds far larger models on classification—which indicates that the prevailing emphasis on scale is partly misallocated for deployed TSC. Convolutional, MLP-mixer, and (prospectively) state-space backbones are well suited to this setting, given their favorable latency relative to attention.

7 Discussion: Gaps and Practical Guidance

Examining the four groups in light of the populated spine and survey (Figure 2, Table 2) yields several observations.

The field is concentrated in time-series-native pre-training. All seven catalogued pre-training objectives are represented natively, and time-series-native is the only branch with architectural diversity; the cross-modal branches, though increasingly populated, each collapse to a single Transformer architecture and span only one or two objectives. The cross-product of modality and objective remains sparse.

Near-uniform reliance on Transformer architectures, except at the edge. The cloud tier is almost entirely Transformer; the non-Transformer models cluster at the lightweight end—convolutional contrastive, transformation-prediction, and self-distillation encoders, the recurrent Activity2Vec, and the MLP-mixers TTM [31] and TSPulse [30]. State-space backbones, despite their linear-time inference and suitability for edge deployment, remain essentially absent from *classification*-oriented FMs, which is the clearest open architectural opportunity.

Adaptation, rather than pre-training, is the principal bottleneck for TSC. Many of the strongest models expose only a forecasting interface; using them for classification means probing or fine-tuning transferred representations. True zero-shot classification—frozen encoder plus a label decision with no task-specific training—remains rare and under-evaluated, even though it is the regime that the label scarcity of TSC most requires. HARGPT [19], which prompts a frozen LLM to recognize activities with no training, and PlotPrompt [39], which prompts a frozen multimodal model with a plot of the signal, are among the few examples.

Channel and length handling are under-served. Several headline models are univariate-only, yet the UEA benchmark is multivariate and unequal-length, and real signals are often irregularly sampled. The models that explicitly target this shape (e.g. ViTST [28]) come disproportionately from the vision-transferred branch.

The edge tier is small but expanding, and need not compromise accuracy. Much deployed TSC—wearable ECG, on-device activity recognition—requires small, low-latency models. This tier (MANTIS [29], TTM [31], TSPulse [30]) is still sparser than the cloud-scale forecasters, but TSPulse's result—state-of-the-art multivariate classification from 1M parameters, exceeding models 10–100$\times$ larger—indicates that the prevailing emphasis on scale is partly misallocated for TSC.

Post-training and alignment are nascent for

Scenario	Viable region of the taxonomy
Wearable, multivariate, few labels	*Design:* contrastive, self-distillation, or dual-space masked. *Scale:* S (∼1M). *Interface:* frozen encoder with linear probe or a light fine-tune head. *Deployment:* GPU-free edge (e.g. TSPulse, MANTIS, TTM).
Cloud, univariate, no labels	*Design:* large autoregressive or masked native models. *Scale:* L. *Interface:* zero-shot via transferred representations and a nearest-centroid decision. *Deployment:* cloud GPU.
Irregular / mixed-rate, moderate labels	*Design:* vision-transferred (image rendering tolerates irregular sampling) or any-variate masked models. *Interface:* fine-tuning, with channel/length handling the deciding constraint. *Deployment:* cloud or edge by model choice.

Table 3: Mapping deployment constraints onto the taxonomy. The objective is not to prescribe specific models but to identify the *region*: a system-driven taxonomy enables a practitioner to narrow the space to a family before comparing individual models.

time series. The developer-side stage that follows pre-training in modern LLMs—instruction tuning, preference optimization—has barely been explored for time-series FMs, where it typically collapses into ordinary fine-tuning. We deliberately keep this distinct from the user-side adaptation axis.

Evaluation is fragmented. There is no widely adopted benchmark that measures foundation models *as classifiers* across UCR/UEA in zero-shot, few-shot, and fine-tuned regimes under a common protocol. Controlled comparisons exist only in narrower settings—Ek et al. [50], for instance, find that a masked-autoencoder objective outperforms contrastive methods for wearable HAR under a shared protocol—but no equivalent exists at the breadth of UCR/UEA. This makes cross-model claims difficult to compare and is, in our assessment, the most actionable gap for the field.

The taxonomy is prescriptive, not only diagnostic. Beyond exposing these gaps, the same organization guides model selection: because its axes are the system constraints a practitioner controls—input dimensionality, label budget, and hardware target—deployment requirements map directly onto regions of it. Table 3 illustrates this for three common scenarios, narrowing the space to a family before individual models are compared.

8 Related Work

The two closest surveys are Liang et al. [2], a broad tutorial-and-survey organized around architecture, pre-training, and modality for time series *analysis*, and Ye et al. [3], which centers the large-language-model route to time series representation. A third survey, Zhang et al. [45], focuses specifically on self-supervised learning for time series, taxonomizing methods as generative, contrastive, and adversarial according to their pre-training objective, while Xu et al. [54] survey the broader multimodal-LLM landscape, organized by the modality into which a series is cast (text, image, graph, audio, table). These surveys are task-broad and built around how models are constructed. Our taxonomy differs in two ways: it is *classification-centric*, foregrounding the adaptation and channel/length axes that decide TSC usability, and it is *system-driven*, organizing the space so that capability and deployment constraints—not merely construction details—are treated as primary. We regard these as complementary: the broad surveys catalog the field, while our framework provides a means of selecting and positioning models for a concrete classification system.

9 Conclusion

We presented a system-driven account of foundation models for time series classification, organized around the four groups that specify any such model—design, scale, interface, and deployment (Figure 1). Within the design group we isolated a well-defined, three-level spine—source modality, pre-training objective, architecture—and used it to place more than forty models. The account is as much diagnostic as descriptive. It reveals a field concentrated in time-series-native, Transformer-based design, with architecturally narrow cross-modal branches; an adaptation bottleneck in the interface that is specific to classification; an emphasis on scale that compact models such as TSPulse suggest is partly misallocated for TSC; a small but expanding edge tier; and the absence of a common benchmark for evaluating these models as classifiers. We anticipate that both the framework and the gaps it identifies will be of use: the former to practitioners selecting a model for a deployed classifier, and the latter to researchers determining where future contributions are most needed.

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